

# **Manufacturing Route Assessment for a Landing Gear Door Hinge**

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## Executive Summary

The aerospace manufacturing sector faces increasing pressure to reduce material waste, environmental impact and lead times, while maintaining high levels of safety and certification compliance. This report assesses the manufacturing route for a landing gear door hinge, examining whether additive manufacturing (AM) is a viable alternative to conventional CNC machining.

The current manufacturing process, based on machining from wrought aluminium or steel alloys, was evaluated alongside metal AM processes. The assessment considered component function and loading, material performance, certification and qualification requirements, environmental impact and implementation considerations. A structured adoption analysis was conducted to compare the strengths, limitations and risks associated with each manufacturing route

The assessment found that conventional machining remains the preferred route for the selected component due to its high certification maturity, predictable material behaviour and established inspection methods. While AM offers advantages in geometric freedom, material efficiency and potential weight reduction, its application to this component is constrained by fatigue performance inconsistency, extensive certification and qualification requirements and practical implementation hardships. Additionally, the need for downstream machining of critical features limits the extent to which AM can fully replace conventional processes.

It is therefore recommended that AM can be adopted selectively, focusing on future design iterations or new aircraft platforms where it can benefit from extended timelines to offset the longer certification process. Conventional machining should be retained for mature, fatigue-loaded components, while organisations maintain readiness to utilise AM as the technology and certification framework continue to mature.

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# 1. Introduction

The aerospace manufacturing sector faces increasing pressure to deliver lighter, higher-performance components, while reducing cost, lead time and environmental impact. Airbus highlights that sustainability is now a core focus in their business strategy in their 2024 Annual Report, with Boeing considering sustainability standards a risk factor for business operations (Airbus SE, 2025; The Boeing Company, 2025). While conventional manufacturing approaches remain reliable and tested, they can, in some cases, result in higher material waste and limit design freedom. To keep up with increased performance and sustainability constraints, advanced manufacturing alternatives are being considered.

This report focuses on the manufacture of a landing gear door hinge. Traditionally, this part is machined from high-strength alloys. This is a proven and certified technique but entails material inefficiency and design constraints due to tooling access. As a result of these constraints, the component is a suitable case for evaluating alternative advanced manufacturing approaches.

This report will assess the feasibility of an alternative manufacturing approach. The evaluation considers multiple factors, not technical performance alone. These include technical feasibility, cost, sustainability considerations and manufacturing maturity and associated risk. This assessment will be used to guide an informed decision regarding potential adoption of the alternative approach.

The aim of this report is to provide a structured comparison between conventional and alternative manufacturing approaches for the landing gear door hinge. The scope is limited to manufacturing considerations, instead of a detailed structural redesign. The report concludes with a recommendation on the suitability of adoption and identifies key constraints and drivers for manufacturing change.

## 2. Background

### 2.1. Component Function and Requirements

The door hinge provides a pivot point for the landing gear door to rotate between the open and closed positions during landing gear extension and retraction. During operation, the hinge needs to be able to withstand significant aerodynamic loads and inertial loads. Failure of the hinge does not result in a flight-critical condition, as continued safe flight and landing remains possible. However, this failure results in reduced aerodynamic and operational margins and is therefore addressed through fail-safe and damage-tolerant design.

### 2.2. Conventional Manufacturing Route

Typically, the hinge would be manufactured using CNC machining from aluminium or steel alloys (Udash, n.d.). The raw material would likely be supplied in wrought form; this would be chosen over casting due to increased fatigue performance. This route offers high dimensional accuracy and well-characterised fatigue resistance, which is required for a component that is loaded cyclically throughout the service life of the aircraft. The maturity of established manufacturing and inspection methods supports the need for fail-safe and damage-tolerant design. However, the route typically results in high buy-to-fly ratio and limits geometric and design optimisation (Groover, 2021; Watson and Taminger, 2018).

### 2.3. Additive Manufacturing Route

Additive Manufacturing is a manufacturing method where components are produced layer by layer from CAD models. In aerospace applications, metal AM processes such as powder bed fusion (PBF) and directed energy deposition (DED) are used to manufacture structural and non-structural components. A well-known example of metal AM is the fuel nozzle used in the LEAP 1A/B engines, which is manufactured using laser powder bed fusion (L-PBF) (Blakey-Milner *et al.*, 2021). AM offers near net shape manufacture and increases geometric freedom compared to conventional subtractive methods. This can support topology optimisation and part consolidation, reduce material waste and in some cases reduce component mass. However, AM also presents challenges related to surface roughness, residual stresses and variability in fatigue performance. As AM is a relatively recent technology in aviation, it faces certification challenges, with the European Union Aviation Safety Agency highlighting that AM processes require close control to consistently produce safe parts (European Union Aviation Safety Agency, 2023).

### **3. Manufacturing Route Assessment**

#### **3.1. Rationale for the Conventional Manufacturing Route**

Wrought aluminium and steel alloys have long been used in the aerospace sector, providing predictable material properties, favourable fatigue behaviour and good corrosion resistance. These characteristics are essential for components operating in the cyclic loading and environmental conditions experienced by aircraft structures. The predictability of the materials is crucial, as it reduces uncertainty in the performance of the component throughout the service life of the aircraft.

CNC machining is routinely selected for the manufacture of aerospace components, due to certification maturity, proven reliability and associated risk reduction. Decades of industrial use mean the effects of the process on material properties are well researched and understood. CNC machining can introduce surface-level residual stresses and affect surface integrity; however, these effects are well understood, controllable and have minimal impact on bulk material properties.

As a result, CNC machining within the aerospace industry is supported by established quality management standards – ISO 9001 and aerospace-specific AS9100 standard. The current AS9100 standard (Rev D) was published in 2016, highlighting a stable and well-established certification framework (SAE International, 2016). This stability reduces uncertainty for aircraft and component manufacturers by supporting continued compliance over the aircraft service life. In addition, conventional machined components benefit from matured inspection and non-destructive testing methods, further reducing uncertainty during qualification and in-service monitoring.

#### **3.2. Limitations of the Conventional Approach**

Despite its maturity and reliability, machining presents several environmental challenges. Components manufactured by machining often have low material efficiency due to the large amount of material removal, requiring this material to be recycled, which is an energy extensive process (Watson and Taminger, 2018). Additionally, machining processes demonstrate low overall energy efficiency, as only a small fraction of total machine energy consumption is associated with material removal (Gutowski, Dahmus and Thiriez, 2006). Significant volume of freshwater is used in coolant preparation and machine and part cleaning, generating contaminated wastewater, that require treatment prior to disposal (Zhao, Ogaldez and Sutherland, 2012). Furthermore, conventional cutting fluids are often chemically hazardous, requiring controlled handling and disposal under regulatory framework (Dahmus and Gutowski, 2004). However, the environmental impacts of machining processes are well characterised, with mature mitigation strategies such as coolant recycling and process optimisation, which can reduce their overall environmental burden.

CNC machining is limited in its ability to achieve high geometric material efficiency, as complex features and weight-optimised geometries are often infeasible, without multiple machining stages and excessive material removal, further extending the overall environmental footprint of the manufacturing stage.

#### **3.3. Motivation for Considering Additive Manufacturing**

These limitations motivate the consideration of an alternative manufacturing route, particularly additive manufacturing. Unlike subtractive manufacturing, additive manufacturing can achieve high material efficiency, with a significantly larger portion of the starting material being used in the final part. This offers the potential to significantly reduce material waste and improve buy-to-fly ratios. AM typically does not require the use of cutting fluids, meaning there are fewer pollutants associated with the process. Additionally, the design freedom offered by additive manufacturing allows for the manufacture of complex internal features and weight-optimised geometries, that would be difficult and required multiple stages to produce using machining. From a manufacturing perspective, additive manufacturing has the potential to reduce process lead time, associated energy consumption and related environmental impacts. However, it should be noted that the environmental performance of AM is highly application dependent and for a small or geometrically simple component, the energy consumption associated with the AM process may exceed that of conventional machining routes (Watson and Taminger, 2018).

## **4. Feasibility of Additive Manufacturing for the Selected Component**

### **4.1. Geometric and Design Feasibility**

From a geometric perspective AM offers capabilities not achievable using conventional machining processes. Unlike machining, AM processes are not limited by tool access, enabling the manufacture of complex internal features, weight-optimised geometries. This design freedom supports topology optimisation, allowing material to be placed more efficiently along load paths. For the selected component, this design freedom can reduce component weight and functional integration that would not be possible with a machined part. However, AM also introduces geometric constraints related to build orientation, overhang angles and the need for support structures, which can increase material wastage. Overall, while AM shows clear geometric advantages, the feasibility of these benefits is highly dependent on component design and process selection.

### **4.2. Material and Performance Considerations**

From a material performance perspective, fatigue behaviour is a critical consideration for the selected component, which is subjected to repeated cyclic loading throughout its life in service. Metal AM processes can introduce defects such as porosity and lack of fusion, as well as greater microstructure variability compared to wrought materials, which can negatively impact fatigue performance (Javidrad *et al.*, 2024). Additionally, as-built AM surfaces exhibit higher surface roughness, introducing local stresses and providing sites for crack initiation (Javidrad *et al.*, 2024). The layer-wise nature of AM can introduce residual stresses, which can further influence fatigue life (Javidrad *et al.*, 2024). To mitigate these effects, AM components often require post processing steps to improve material consistency and surface condition (Javidrad *et al.*, 2024). As a result, achieving consistent fatigue performance in AM components is highly dependent on process control and post-processing, introducing greater uncertainty compared to machined wrought materials.

### **4.3. Certification and Qualification Implications**

From a certification perspective, AM presents greater qualification challenges than conventional machining for the selected component. Unlike machining from wrought material, which benefits from high certification maturity and long service history, metal AM processes exhibit greater variability. As a result, qualification of AM components typically requires extensive validation at the material, process and part level to demonstrate consistent mechanical performance.

Additionally, inspection and non-destructive testing of AM components can be more complex, particularly where internal features are present, as defect populations are highly dependent on process parameters. Regulators therefore place strong emphasis on process control, documentation, and traceability to ensure repeatable outcomes. The European Union Aviation Safety Agency highlights that additive manufacturing requires close control of the manufacturing process to consistently produce safe and compliant parts (European Union Aviation Safety Agency, 2025). As a result, manufacturers should expect increased regulatory involvement and audit activity, reflecting the sensitivity of the technology. Furthermore, certification of AM components can take significantly longer compared to conventional machining routes (Zhang and Liang, 2019).

## 5. Adoption Analysis

### 5.1. SWOT Analysis for Additive Manufacturing Adoption

Table 5.1: SWOT analysis for additive manufacturing adoption

| Category             | Key Points                                                                                                                                                                                                                                                                    |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Strengths</b>     | <ul style="list-style-type: none"><li>• High material efficiency and reduced buy-to-fly ratio</li><li>• Increased geometric freedom enabling internal features and topology optimisation</li><li>• Potential for part consolidation and reduced assembly complexity</li></ul> |
| <b>Weaknesses</b>    | <ul style="list-style-type: none"><li>• Variable fatigue performance compared to wrought material</li><li>• High surface roughness requiring post-processing</li><li>• Lower certification maturity and limited-service history</li></ul>                                     |
| <b>Opportunities</b> | <ul style="list-style-type: none"><li>• Alignment with aerospace sustainability and waste-reduction targets</li><li>• Suitability for design iterations</li><li>• Increasing regulatory guidance and industrial experience with metal AM</li></ul>                            |
| <b>Threats</b>       | <ul style="list-style-type: none"><li>• High qualification and certification cost</li><li>• Extended certification timelines and programme risk</li><li>• Limited aerospace-qualified AM supply chain</li></ul>                                                               |

#### Adoption Analysis

### 5.2. SWOT Analysis for Additive Manufacturing Adoption

Table 5.1 summarises the key strengths, weaknesses, opportunities and threats assigned with additive manufacturing adoption for the selected component.

### 5.3. Implementation Considerations

From an implementation perspective, adoption of AM would require significant upfront investment, including specialised equipment, facilities and workforce training. In addition to hardware costs, manufacturers must account for the time and resources required for process development, validation and certification to demonstrate consistent and repeatable material performance. This qualification effort can be significantly greater than conventional routes, extending development timelines and increasing programme risk. Supply chain readiness also presents a challenge, as the availability of aerospace-qualified AM suppliers and established material specifications remain limited compared to conventional routes. Additionally, some aircraft components, such as a landing gear door hinge would still require the use of precision machining, to create holes for attachment bolts, resulting in a hybrid manufacturing route rather than a full replacement of conventional machining. As a result, implementation of AM is more suited to future design iterations or new aircraft platforms, where longer development timelines can be accommodated rather than direct replacement of established machined components within mature aircraft programmes.

In the longer term, with the development of Industry 4.0, emerging digital twin technologies may support the implementation and certification of AM components. Digital twins can be used to simulate AM process behaviour and predict material properties by linking process parameters to microstructural development and performance (Zhao, Vieu, and Siemens, 2025; Zhang *et al.*, 2020). For metal AM, this capability has the potential to improve process understanding, reduce variability and provide supplementary evidence during qualification activities. While such tools are not fully developed or accepted as a replacement for physical testing, they may help reduce the overall certification burden by supporting process validation and sensitivity analysis. As digital twin maturity and regulatory confidence increase, their implementation into the AM workflow could improve feasibility of certification for future aerospace applications.

## 6. Strategic Recommendations

- Adopt additive manufacturing selectively, focusing on future design iterations or new aircraft platforms where longer development timelines and qualification effort can be accommodated.
- Retain conventional CNC machining for mature, fatigue-loaded components where certification maturity, reliability and predictability remain vital.
- Use additive manufacturing only where it delivers clear functional or system level benefits, such as part consolidation or weight reduction that cannot be achieved with machining alone.
- Plan for hybrid manufacturing routes, recognising that precision machining will still be required for critical interfaces and tolerances, limiting a full process replacement
- Engage early with regulators to reduce certification risk and align process development with regulatory expectations.
- Maintain organisational readiness for additive manufacturing, by developing internal capabilities and process knowledge so that adoption can be accelerated as the technology and certification framework mature.

## 7. Conclusion

This report assessed the manufacturing route for a landing gear door hinge, evaluating the suitability of additive manufacturing as an alternative to conventional CNC machining. While machining from wrought aluminium or steel alloy remains the preferred route due to certification maturity, predictable material performance and established inspection practices, it presents limitations with material efficiency, design flexibility and environmental impact. AM offers clear advantages in geometric freedom and material utilisation, but its adoption for this component is constrained by fatigue performance variability, qualification effort and regulatory issues. As a result, AM is not currently suited as a direct replacement for conventional machining for this application. However, it presents a viable alternative for future design iterations where its benefits can be fully utilised. A selective and preparedness focused strategy is therefore recommended, allowing manufacturers to utilise AM as certification maturity and industry capability continue to develop.



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